

Danmarks Tekniske Universitet

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Course title: Bayesian Machine Learning

Course number: 02477

Aids allowed: All aids except internet and AI

Exam duration: 4 hours

Weighting: 100%

02477 Bayesian Machine Learning Exam 2025

Technical University of Denmark

- **Duration:** 4 hours
- **Aids:** All aids except internet and AI
- **Results:** Report all numeric results with **2 digits** after the decimal point.
- **Explain** how you arrived at your results, document intermediate results when possible.
- **Hand-in:** Your solution can be either digitally as a PDF, on paper or both.
- **Student number:** Make sure your student number is **visible on all pages**.

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Part 1: A linear model for regression

Consider a regression problem of the form

$$y_n = f(x_n) + \epsilon_n, \quad (1)$$

where $y_n \in \mathbb{R}$ is the observed value for the n 'th input $x_n \in \mathbb{R}$ and $\epsilon_n \sim \mathcal{N}(0, \sigma^2)$ is i.i.d. additive noise with variance $\sigma^2 > 0$. The model $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by

$$f(x_n) = w_0 + w_1 x_n + w_2 x_n^2 + w_3 x_n^3 = \mathbf{w}^T \boldsymbol{\phi}(x_n), \quad (2)$$

where the parameter vector $\mathbf{w}$ is defined as $\mathbf{w} = [w_0 \ w_1 \ w_2 \ w_3] \in \mathbb{R}^4$ and $\boldsymbol{\phi}(x_n)$ is a polynomial feature representation defined as $\boldsymbol{\phi}(x_n) = [1 \ x_n^1 \ x_n^2 \ x_n^3] \in \mathbb{R}^4$.

Assume the following i.i.d. prior for each parameter $w_i \sim \mathcal{N}(0, 2)$ and assume $\sigma = \frac{1}{5}$.

Below you are given a dataset of $N = 5$ observations:

$$\mathbf{x} = [0.0 \ 1.0 \ 2.0 \ 4.0 \ 5.0] \quad (3)$$

$$\mathbf{y} = [1.0 \ 0.5 \ -0.1 \ -0.9 \ 1.1] \quad (4)$$

such that the n 'th entry in $\mathbf{x}$ is x_n and similar for $\mathbf{y}$. Let $\Phi \in \mathbb{R}^{N \times 4}$ denote the corresponding design matrix for the dataset, i.e. the n 'th row of Φ is $\boldsymbol{\phi}(x_n)$.

Question 1.1: Determine the analytical distribution $p(\mathbf{y}|\mathbf{w}, \Phi, \sigma^2)$.

Question 1.2: Compute the maximum likelihood estimator for $\mathbf{w}$.

Question 1.3: Determine the analytical distribution $p(\mathbf{w}|\mathbf{y})$.

Question 1.4: Determine the analytical distribution $p(y^*|\mathbf{y}, x^*)$ for $x^* = 3$.

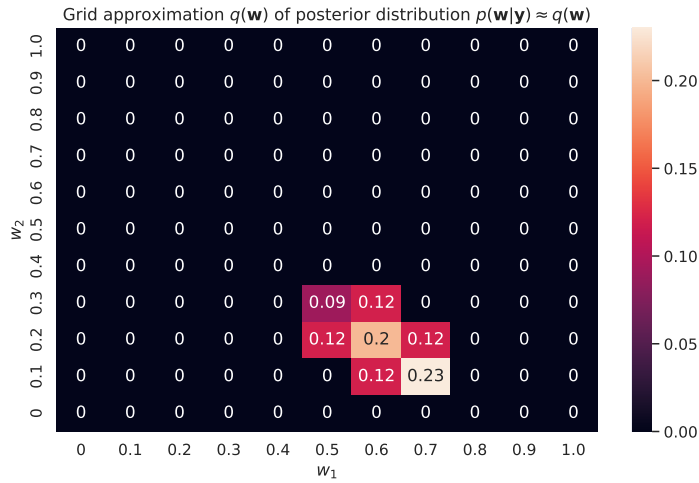


Figure 1: Grid approximation $q(\mathbf{w})$ for $p(\mathbf{w}|\mathbf{y}) \approx q(\mathbf{w})$, where $\mathbf{w} = [w_1 \ w_2] \in \mathbb{R}^2$. Each cell in the plot shows the approximate posterior probability for each pair of (w_1, w_2) , i.e. $q(w_1 = 0.7, w_2 = 0.1) = 0.23$. The approximation $q(\mathbf{w})$ is zero for all values of $\mathbf{w} = (w_1, w_2)$ not listed explicitly in the plot.

Part 2: Grid approximation for non-linear regression

Consider a probabilistic model with the following likelihood

$$p(y_n|\mathbf{w}, x_n) = \mathcal{N}(y_n|f(x_n), \sigma^2), \quad (5)$$

where $f(x_n) = e^{w_1 + w_2 x_n}$ and $\sigma^2 = \frac{1}{2}$ is the noise variance. Figure 1 shows a (discrete) grid approximation $q(\mathbf{w})$ for the posterior distribution $p(\mathbf{w}|\mathbf{y})$ such that $p(\mathbf{w}|\mathbf{y}) \approx q(\mathbf{w})$, where $\mathbf{y}$ is a vector of observations (not given).

Question 2.1: Use the grid approximation to estimate the posterior mean and variance of w_1 .

Question 2.2: Use the grid approximation to estimate $p(y^*|\mathbf{y}, x^*)$ for $x^* = 4$ and $y^* = 4$.

Part 3: A mixture model

Consider the following likelihood for $y_n \in \mathbb{R}$

$$p(y_n|\boldsymbol{\theta}, \mathbf{x}_n, z_n) = (1 - z_n)\mathcal{N}(y_n|0, 1) + z_n\mathcal{N}(y_n|\boldsymbol{\theta}^T \mathbf{x}_n, \sigma^2), \quad (6)$$

where $\boldsymbol{\theta} = [\theta_1 \ \theta_2] \in \mathbb{R}^2$ is a parameter vector to be estimated, $z_n \in \{0, 1\}$ is a binary indicator variable, and $\mathbf{x}_n \in \mathbb{R}^2$ is the input feature for the n 'th observation. The noise variance is $\sigma^2 = 1/10$.

Assume the following prior distributions

$$\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1} \mathbf{I}) \quad (7)$$

$$z_n \sim \text{Ber}(z_n|\pi), \quad (8)$$

where $\pi = \frac{4}{5}$ and $\alpha = \frac{1}{2}$.

Suppose we make a single observation, i.e. $N = 1$, where $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $y_1 = 1$. The resulting joint distribution is

$$p(y_1, \boldsymbol{\theta}, z_1) = [(1 - z_1)\mathcal{N}(y_1|0, 1) + z_1\mathcal{N}(y_1|\boldsymbol{\theta}^T \mathbf{x}_1, \sigma^2)] \text{Ber}(z_1|\pi) \mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \alpha^{-1} \mathbf{I}). \quad (9)$$

Question 3.1: Determine the distribution $p(y_1, \boldsymbol{\theta})$ analytically.

Question 3.2: Determine the distribution $p(y_1, z_1)$ analytically.

Let $\mathcal{Q}_1$ be the mean-field Gaussian family and $q_1^*(\boldsymbol{\theta}) = \mathcal{N}(\theta_1|m_1, v_1)\mathcal{N}(\theta_2|m_2, v_2)$ be the optimal variational approximation such that

$$q_1^*(\boldsymbol{\theta}) = \arg \min_{q \in \mathcal{Q}_1} \text{KL}[q(\boldsymbol{\theta})||p(\boldsymbol{\theta}|y_1)] \quad (10)$$

for

$$m_1 = m_2 = \frac{1}{4}, \quad v_1 = v_2 = \frac{6}{5}. \quad (11)$$

Question 3.3: Compute the entropy of $q_1^*(\boldsymbol{\theta})$.

Question 3.4: Compute a 95% interval for $q_1^*(\theta_2)$.

Question 3.5: Determine the approximate predictive distribution $p(y^*|\mathbf{y}, \mathbf{x}^*)$ analytically for $\mathbf{x}^* = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ using the approximation $p(\boldsymbol{\theta}|\mathbf{y}) \approx q_1^*(\boldsymbol{\theta})$.

Let $\mathcal{Q}_2$ be the family of full-rank Gaussians. Suppose you are unhappy with the predictive performance that you obtained using $q_1^*(\boldsymbol{\theta})$, and you decide to change the variational family to $\mathcal{Q}_2$ and update your variational approximation such that

$$q_2^*(\boldsymbol{\theta}) = \arg \min_{q \in \mathcal{Q}_2} \text{KL}[q(\boldsymbol{\theta})||p(\boldsymbol{\theta}|y_1)] \quad (12)$$

Question 3.6: Explain how you expect the value of $\text{KL}[q_1^*(\boldsymbol{\theta})||p(\boldsymbol{\theta}|y_1)]$ to change when you switch from $q_1^*(\boldsymbol{\theta})$ to $q_2^*(\boldsymbol{\theta})$. Explain your reasoning.

Part 4: Gaussian process classification

Consider a Gaussian process classification model given by

$$f(x) \sim \mathcal{GP}(0, k_1(x, x')), \quad (13)$$

$$y_n | f_n \sim \text{Ber}(\sigma(f(x_n))), \quad (14)$$

where $\sigma : \mathbb{R} \rightarrow (0, 1)$ is the logistic sigmoid function, $f_n = f(x_n)$, and the covariance function k_1 is given by

$$k_1(x, x') = 5 \left(1 + \exp \left(-\frac{1}{4}(x - x')^2 \right) \right). \quad (15)$$

Let $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^N$ denote a dataset with $N = 3$ observations given by

$$\mathbf{x} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \quad (16)$$

$$\mathbf{y} = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}. \quad (17)$$

Let $\mathbf{f} = [f(x_1) \ f(x_2) \ f(x_3)]$ and let $q(\mathbf{f})$ denote the Laplace approximation for the posterior distribution of $\mathbf{f}$ such that $p(\mathbf{f}|\mathbf{y}) \approx q(\mathbf{f}) = \mathcal{N}(\mathbf{f}|\mathbf{m}, \mathbf{S})$, where

$$\mathbf{m} = \begin{bmatrix} 1.56 \\ 0.84 \\ -0.32 \end{bmatrix} \quad \mathbf{S} = \begin{bmatrix} 2.69 & 1.46 & 0.42 \\ 1.46 & 1.78 & 1.2 \\ 0.42 & 1.2 & 2.13 \end{bmatrix}. \quad (18)$$

Finally, let $f^* = f(x^*)$ and $x^* = 4$.

Question 4.1: Determine the analytical prior distribution for $p(f^*)$

Question 4.2: Compute the prior predictive probability $p(y^* = 0|x^*)$.

Question 4.3: Compute the (approximate) posterior distribution $p(f^*|\mathbf{y}, x^*)$ using the Laplace approximation $q(\mathbf{f})$.

If you did not solve Question 4.3, you can assume $p(f^*|\mathbf{y}, x^*) = \mathcal{N}(f^* | -1, 11)$ when solving Question 4.4.

Question 4.4: Compute the (approximate) posterior predictive distribution $p(y^*|\mathbf{y}, x^*)$ using the probit approximation.

Let $k_2(x, x')$ be another kernel function given by $k_2(x, x') = \frac{1}{10}k_1(x, x')$.

Question 4.5: Explain how you expect the (approximate) posterior predictive probability $p(y^* = 1|\mathbf{y}, x^*)$ to change if the covariance function is switched from k_1 to k_2 . Explain your reasoning.

Part 5: MCMC sampling

Consider the following probabilistic model:

$$x|z \sim \mathcal{N}(0, e^z), \quad (19)$$

$$z \sim \mathcal{N}(z|1, 1). \quad (20)$$

Question 5.1: Implement a Metropolis-sampler for the distribution $p(z, x)$. Use a Gaussian proposal distribution with covariance matrix $2I$, where I is the identity matrix. Run two chains initialized as $(z, x) = (-1, 0)$ and $(z, x) = (2, 5)$. Run 3000 iterations and do not discard any warm up samples. Plot the traces for both chains and report the acceptance rates.

Use the MCMC samples of (z, x) from Question 5.1 to answer the next two questions. If you did not solve Question 5.1, you can draw 6000 samples of (z, x) as

$$\begin{bmatrix} z \\ x \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.5 \\ 0.5 & 5 \end{bmatrix}\right) \quad (21)$$

and assume these are samples from $p(z, x)$ when solving the next two questions.

Question 5.2: Estimate the mean of $f(z, x) = (z - x)^2$.

Question 5.3: Estimate the probability of the event $x < 1$.